

AN ORDERED PAIR OF GRAPHS OF NONZERO TWIN-WIDTH ATTAINING THE PETTERSSON–SYLVESTER TENSOR-PRODUCT BOUND

ABSTRACT. Pettersson and Sylvester proved that

$$\text{tw}(G \times H) \leq \max\{\text{tw}(G)\Delta(H) + \Delta(H), \text{tw}(H)\} + \Delta(H)$$

for the tensor (direct) product of any two graphs, and wrote in their Conclusion that it would be nice to find two graphs of *nonzero* twin-width for which this bound is tight. We exhibit such a pair. Let $G = C_8(1, 2)$ be the square of the 8-cycle and let $H = C_5$; both have twin-width 2. Then

$$\text{tw}(C_8(1, 2) \times C_5) = 8 = \max\{2 \cdot 2 + 2, 2\} + 2,$$

and more generally $\text{tw}(C_8(1, 2) \times C_n) = 8$ for every $n \geq 5$. The bound is not symmetric in (G, H) , and the claim is for this ordering only. The lower bound is a three-case count over the $\binom{40}{2} = 780$ vertex pairs of the 40-vertex product, which is printed here as a graph6 string.

1. THE QUESTION

All graphs are finite, simple and undirected. We use the contraction-sequence definition of twin-width of Bonnet, Kim, Thomassé and Watrigant [2]: a *contraction sequence* of an n -vertex graph X is a sequence of partitions of $V(X)$ starting at the partition into singletons and ending at the one-part partition, each obtained from the previous one by merging two parts. A pair of distinct parts P, Q is *black* if every $p \in P, q \in Q$ is an edge, *red* if some such pair is an edge and some is not, and *absent* if none is. The *width* of the sequence is the maximum, over all partitions occurring in it (the initial one included), of the maximum red degree of a part; $\text{tw}(X)$ is the least width of a contraction sequence of X . Thus $\text{tw}(X) = 0$ if and only if X is a cograph.

The *tensor product* (also direct, Kronecker, weak or conjunctive product) $G \times H$ has vertex set $V(G) \times V(H)$, with $(u, i)(v, j)$ an edge if and only if $uv \in E(G)$ and $ij \in E(H)$. Consequently

$$(1) \quad N_{G \times H}((u, i)) = N_G(u) \times N_H(i),$$

so $G \times H$ is $\Delta(G)\Delta(H)$ -regular whenever G and H are regular.

2020 *Mathematics Subject Classification.* 05C76, 05C75, 68R10.

Key words and phrases. twin-width, tensor product, direct product, circulant graph, contraction sequence, tightness.

Pettersson and Sylvester [4] proved, for arbitrary graphs G and H and with no hypothesis beyond simplicity,

$$(2) \quad \text{tww}(G \times H) \leq \max\{\text{tww}(G)\Delta(H) + \Delta(H), \text{tww}(H)\} + \Delta(H).$$

The first of the two questions in the unnumbered Conclusion of [4] reads, verbatim:

Question. It would be nice to find an example of two graphs G, H with non-zero twin-width for which Theorem [the bound (2) above] is tight.

The bracketed text replaces a cross-reference left unresolved in the source; nothing else in the sentence is altered. We read “tight” as equality in (2) at one ordered instantiation, which is what is exhibited below; a reading that demands more — equality for both orderings of the pair, say — is not met here. The tight families for (2) given in [4] — products with a complete graph, and Hamming graphs — each have a factor of twin-width 0.

2. THE RESULT

Let $G = C_8(1, 2)$ be the circulant graph on $\mathbb{Z}_8$ in which $u \sim v$ if and only if $u - v \equiv \pm 1$ or $\pm 2 \pmod{8}$; equivalently G is the square of the 8-cycle. It is 4-regular with 16 edges. Let $H = C_5$.

Theorem 1. $\text{tww}(C_8(1, 2)) = \text{tww}(C_5) = 2$, and

$$\begin{aligned} \text{tww}(C_8(1, 2) \times C_5) &= 8 = \max\{\text{tww}(G)\Delta(H) + \Delta(H), \text{tww}(H)\} + \Delta(H) \\ &= \max\{2 \cdot 2 + 2, 2\} + 2. \end{aligned}$$

Corollary 2. $\text{tww}(C_8(1, 2) \times C_n) = 8 = \max\{2 \cdot 2 + 2, 2\} + 2$ for every $n \geq 5$.

The active branch of the maximum is $\text{tww}(G)\Delta(H) + \Delta(H) = 6 > \text{tww}(H) = 2$, so the term attained is the one that carries $\text{tww}(G) \neq 0$.

The only tool needed for the lower bounds is [1, Lemma 3.1], which is quoted in [4] as well.

Lemma 3 (Ahn, Hendrey, Kim, Oum [1]). *For every graph X with at least two vertices,*

$$\text{tww}(X) \geq b_{\min}(X) := \min_{u \neq v} |(N(u) \triangle N(v)) \setminus \{u, v\}|.$$

We write $b(u, v) = |(N(u) \triangle N(v)) \setminus \{u, v\}|$ and $s(u, v) = |N(u) \triangle N(v)|$, and $b_{\min}$, $s_{\min}$ for their minima over unordered pairs of distinct vertices. Lemma 3 bounds tww below by the *minimum*, so exhibiting one pair with a small b proves nothing: the whole profile must be computed.

The two factors. In C_5 we have $b(u, v) = 2$ for every one of the 10 pairs (adjacent pairs have disjoint neighbourhoods, so $s = 4$ and two of the four elements are u, v themselves; pairs at distance 2 have $s = 2$ and neither u nor v lies in the symmetric difference). Hence $b_{\min}(C_5) = 2$ and $\text{tw}(C_5) \geq 2$. For the matching upper bound, contract 0 with 2, then 3 with 4, then the part $\{0, 2\}$ with 1, then the two remaining parts: the maximum red degree never exceeds 2. So $\text{tw}(C_5) = 2$.

In $G = C_8(1, 2)$, with $N(u) = \{u \pm 1, u \pm 2\}$, the quantities s and b depend only on the difference $d = v - u \in \{1, 2, 3, 4\}$:

d	1	2	3	4
$s(u, u + d)$	4	6	4	4
$b(u, u + d)$	2	4	4	4

(For $d = 1$: $N(0) = \{1, 2, 6, 7\}$ and $N(1) = \{0, 2, 3, 7\}$ have symmetric difference $\{0, 1, 3, 6\}$, from which 0 and 1 are deleted.) Hence $s_{\min}(G) = 4$ and $b_{\min}(G) = 2$, so $\text{tw}(G) \geq 2$; and the contraction sequence merging 0 with 1, 2 with 3, 4 with 5, 6 with 7, then $\{0, 1\}$ with $\{4, 5\}$, then $\{2, 3\}$ with $\{6, 7\}$, then the two remaining parts, has width 2. So $\text{tw}(G) = 2$. Both factors have nonzero twin-width, as the question demands.

The product. Put $P = G \times H$. By (1) it has 40 vertices, is 8-regular, and so has 160 edges. The upper bound $\text{tw}(P) \leq 8$ is (2) itself, so all the content is in the lower bound, which we now prove by Lemma 3.

Proof that $\text{tw}(P) \geq 8$. Let $(u, i) \neq (v, j)$ and put $m = |N_G(u) \cap N_G(v)|$ and $m' = |N_H(i) \cap N_H(j)|$. From the profile above, $m \leq 2$ (indeed $m = (8 - s(u, v))/2$), and in C_5 we have $m' = 0$ for adjacent i, j and $m' = 1$ for i, j at distance 2. By (1), $|N_P((u, i))| = |N_P((v, j))| = 8$. Three cases.

(A) $u = v, i \neq j$. Here $N_P((u, i)) \triangle N_P((v, j)) = N_G(u) \times (N_H(i) \triangle N_H(j))$, and neither (u, i) nor (u, j) lies in this set, since $u \notin N_G(u)$; so nothing is deleted and $b = 4|N_H(i) \triangle N_H(j)| \geq 4 \cdot 2 = 8$, with equality exactly when i, j are at distance 2 in C_5 .

(B) $i = j, u \neq v$. Symmetrically $b = 2|N_G(u) \triangle N_G(v)| = 2s(u, v) \geq 2 \cdot 4 = 8$, with equality exactly when $uv \in E(G)$.

(C) $u \neq v$ and $i \neq j$. Then $|N_P((u, i)) \triangle N_P((v, j))| = 8 + 8 - 2|N_G(u) \cap N_G(v)| \cdot |N_H(i) \cap N_H(j)| = 16 - 2mm'$. Moreover (u, i) lies in the symmetric difference if and only if $uv \in E(G)$ and $ij \in E(H)$, and likewise for (v, j) ; so at most 2 is deleted, and only when $uv \in E(G)$ and $ij \in E(H)$. If $ij \in E(H) = E(C_5)$ then $m' = 0$ because C_5 is triangle-free, and $b \geq 16 - 0 - 2 = 14$. Otherwise nothing is deleted and $b = 16 - 2mm' \geq 16 - 2 \cdot 2 \cdot 1 = 12$.

So $b_{\min}(P) = 8$. It is attained in case (A), for instance at the pair $\{(0, 0), (0, 2)\}$, and in case (B), for instance at $\{(0, 0), (1, 0)\}$; and Lemma 3 then gives $\text{tw}(P) \geq 8$. $\square$

Remark 4 (the upper bound without (2)). Taking $\text{tw}(P) \leq 8$ from (2) is legitimate — tightness of a bound *is* the assertion that the bound is attained — but it leaves the equality resting on the statement being tested. It need

not. The accompanying program builds an explicit contraction sequence of the 40-vertex product, 39 merges long, and re-certifies its width as 8 from the original adjacency; that sequence is printed in the recorded run. So $\text{tw}(P) \leq 8$ holds independently of (2), and with $b_{\min}(P) = 8$ it brackets $\text{tw}(P) = 8$ without (2) entering the argument. The same holds for each C_n , $n = 5, \dots, 12$.

This proves Theorem 1. For Corollary 2, replace C_5 by C_n with $n \geq 5$. Then $\Delta(C_n) = 2$ and $\text{tw}(C_n) = 2$: the lower bound is $b_{\min}(C_n) = 2$ exactly as for C_5 (adjacent pairs give $s = 4$ and $b = 2$; pairs at distance 2 give $s = b = 2$; pairs at distance ≥ 3 give $s = b = 4$), and the upper bound is the round-the-cycle sequence, which merges 0 with 1 and then absorbs $2, 3, \dots, n-1$ one at a time and has width 2. So the right-hand side of (2) is again $\max\{2 \cdot 2 + 2, 2\} + 2 = 8$. In the three-case count, (A) gives $4|N_{C_n}(i) \triangle N_{C_n}(j)| \geq 8$, with equality at distance-2 pairs; (B) gives $2s(u, v) \geq 8$ unchanged; and in (C) still $m \leq 2$, while $m' \leq 1$ with $m' = 0$ whenever $ij \in E(C_n)$ (true for $n \geq 5$, as C_n is then triangle-free), so $b \geq 12$. Hence $b_{\min}(C_8(1, 2) \times C_n) = 8$ and $\text{tw}(C_8(1, 2) \times C_n) = 8$. The value $n = 4$ is excluded because C_4 is a cograph, so $\text{tw}(C_4) = 0$ and C_4 is not an admissible factor here.

Scope. Neither half of the certificate is new: the upper half is (2), which is the theorem of [4], the lower half is [1, Lemma 3.1], and the value $\text{tw}(C_n) = 2$ used for the family is also in [1]. We prove no new lemma; the content here is the pair together with the three-case count. Nothing beyond equality in (2) at the ordered pairs $(C_8(1, 2), C_5)$ and $(C_8(1, 2), C_n)$, $n \geq 5$, is claimed: with the roles exchanged the bound reads $\max\{2 \cdot 4 + 4, 2\} + 4 = 16$ and is not attained by the same product; no minimality is asserted, neither of the order 40 nor in any other sense; and we do not claim that no better general bound of the same shape exists. We looked for an earlier example of this kind in the nine works recorded as citing [4] by OpenAlex and by Semantic Scholar as of 29 August 2026, in the DBLP title index for twin-width, and in arXiv abstract searches pairing twin-width with the tensor, direct and Kronecker products, and found none; the nearest work found was [3], on graphs whose twin-width equals the bound of Lemma 3, which does not treat products. That search covered titles, abstracts and citation records rather than full texts, so we make no priority claim.

3. THE WITNESS, AND REPRODUCTION

Theorem 1 and Corollary 2 need no computer: the product is determined by the two parameters, and the lower bound is the three-case count above, whose inputs are the two four-entry tables of s and b . For machine reading we also record $P = C_8(1, 2) \times C_5$ explicitly. Label the vertex (u, i) by $k = 5u + i$, so that $u = \lfloor k/5 \rfloor$ and $i = k \bmod 5$ for $k = 0, \dots, 39$. Then P has the graph6 string obtained by concatenating the three lines

```

g?@LA_gcASsgSgISQc?Ch@Q_Ad?Ad?HQ??Ch?Ad??IS?
?Sg?AS_??AS_?Ad???Sg??@Q_??QcAO??dL??@Q`O??S
gI??AdC_?@IOHQ??Chh0??SgSg??ISDI??AdCh??AS_

```

(one line of 131 characters in total, of lengths 44, 44 and 43), whose 160 edges are exactly the pairs $\{5u + i, 5v + j\}$ with $u - v \equiv \pm 1, \pm 2 \pmod{8}$ and $i - j \equiv \pm 1 \pmod{5}$. The only character in the string that is easy to mistranscribe is the single grave accent (ASCII 0x60) on the second line; every other character is a letter, a digit, `?`, `@` or `_`.

Verification. The file `verify.py` re-derives the computational claims above in exact integer and set arithmetic using only the Python standard library, and `verify.output.txt` records a run of it. It decodes the `graph6` string printed here and checks label-for-label that it is the tensor product built from the two parameters; recomputes the full s - and b -profiles of $C_8(1, 2)$ and of C_5 ; re-verifies the two printed contraction sequences at width 2 in a from-scratch trigraph simulator, so that $\text{tw}(C_8(1, 2)) = \text{tw}(C_5) = 2$ exactly with no exhaustive search in the load-bearing path; enumerates all $\binom{40}{2} = 780$ vertex pairs of the product, splits them into the three cases and reports the per-case minima 8, 8, 12; evaluates both orderings of (2), obtaining 8 and 16; builds and re-certifies the width-8 sequence of Remark 4; and repeats both bounds for C_n , $n = 5, \dots, 12$. Its search for a contraction sequence of the 40-vertex product is a heuristic, so it proves $\text{tw}(P) \leq 8$ — the width of what it returns is re-certified from the adjacency, which is what makes that sound — and it does *not* exclude a width-7 sequence; only Lemma 3 does that. Nothing in the proof above needs the program: every input it consumes is printed here.

REFERENCES

- [1] J. Ahn, K. Hendrey, D. Kim, and S. Oum, *Bounds for the twin-width of graphs*, SIAM J. Discrete Math. **36** (2022), no. 3, 2352–2366. doi:10.1137/21M1452834; arXiv:2110.03957.
- [2] É. Bonnet, E. J. Kim, S. Thomassé, and R. Watrigant, *Twin-width I: tractable FO model checking*, J. ACM **69** (2022), no. 1, Art. 3, 46 pp. doi:10.1145/3486655; arXiv:2004.14789.
- [3] F. Heinrich, F. Ihringer, M. Rassmann, and T. Volk, *On the twin-width of near-regular graphs*, Discrete Appl. Math. **379** (2026), 177–193. doi:10.1016/j.dam.2025.07.044; arXiv:2504.02342.
- [4] W. Petterson and J. Sylvester, *Bounds on the twin-width of product graphs*, Discrete Math. Theor. Comput. Sci. **25** (2023), no. 1, Paper No. 18. doi:10.46298/dmtcs.10091; arXiv:2202.11556v4.